Cyberattack Kill Chain Report

Intrusion Scenario and Mitigation via AppArmor

SOC Analyst / Security Engineer

May 22, 2026

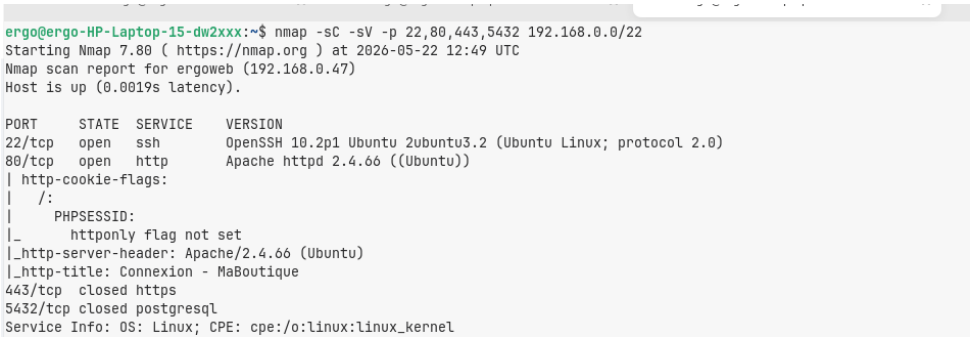
Abstract

This academic report documents a full cyberattack kill chain executed against the SecLab vulnerable environment. It traces the attack from initial reconnaissance using Nmap, to taking control of the server via SQL Injection on the index page, OS command injection on the admin page, followed by a privilege escalation to root using a CopyFAIL-based script. The second phase of the report demonstrates the defensive containment strategy, proving that applying OS-level Mandatory Access Control (AppArmor) effectively neutralizes such attacks.

Contents

1 Phase 1: Reconnaissance and Discovery

The attack commenced with network enumeration to map the target infrastructure. An Nmap scan against the target IP range successfully identified the victim machine at `192.168.0.47`. The scan reported an open HTTP port (80) running Apache httpd.

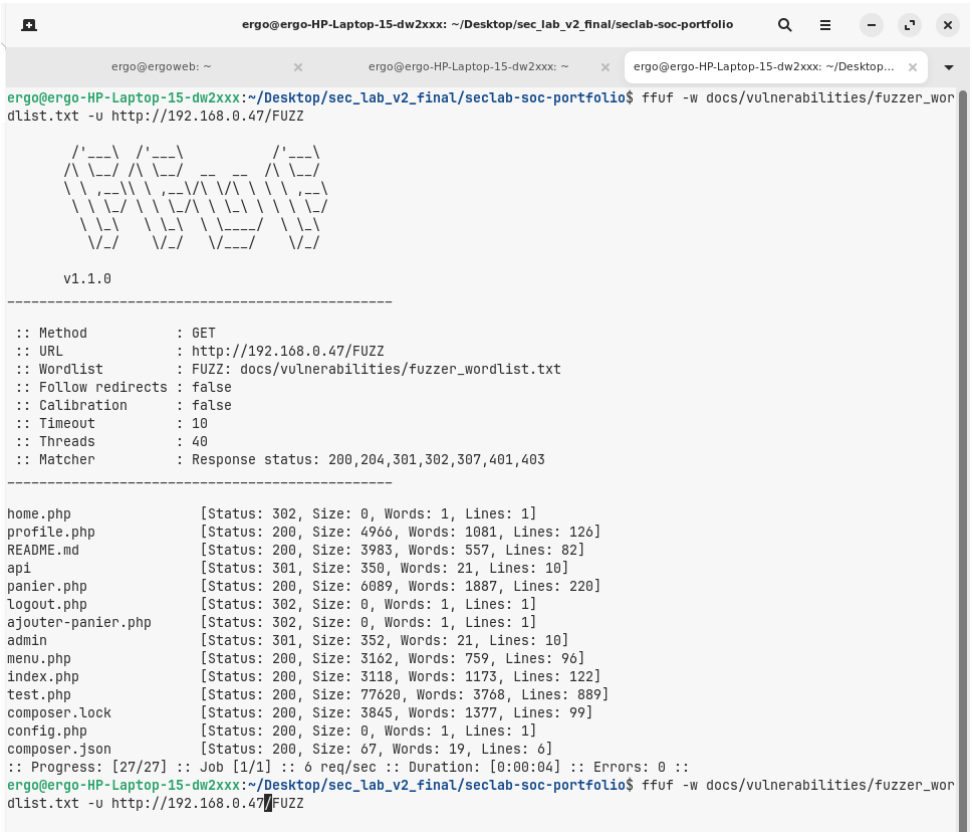


```
ergo@ergo-HP-Laptop-15-dw2xxx:~$ nmap -sC -sV -p 22,80,443,5432 192.168.0.0/22
Starting Nmap 7.80 ( https://nmap.org ) at 2026-05-22 12:49 UTC
Nmap scan report for ergoweb (192.168.0.47)
Host is up (0.0019s latency).

PORT     STATE  SERVICE    VERSION
22/tcp   open   ssh        OpenSSH 10.2p1 Ubuntu 2ubuntu3.2 (Ubuntu Linux; protocol 2.0)
80/tcp   open   http       Apache httpd 2.4.66 ((Ubuntu))
| http-cookie-flags:
|   /:
|     PHPSESSID:
|_      httponly flag not set
|_http-server-header: Apache/2.4.66 (Ubuntu)
|_http-title: Connexion - MaBoutique
443/tcp  closed https
5432/tcp closed postgresql
Service Info: OS: Linux; CPE: cpe:/o:linux:linux_kernel
```

Figure 1: Nmap scan revealing open standard TCP ports, including Port 80.

Following port discovery, the application endpoint map was established using FFUF (Fuzz Faster U Fool) against the discovered HTTP server.



```
ergo@ergo-HP-Laptop-15-dw2xxx: ~/Desktop/sec_lab_v2_final/seclab-soc-portfolio
ergo@ergoweb: ~ | ergo@ergo-HP-Laptop-15-dw2xxx: ~ | ergo@ergo-HP-Laptop-15-dw2xxx: ~/Desktop...
ergo@ergo-HP-Laptop-15-dw2xxx:~/Desktop/sec_lab_v2_final/seclab-soc-portfolio$ ffuf -w docs/vulnerabilities/fuzzer_wor
dlist.txt -u http://192.168.0.47/FUZZ

       /'___\  /'___\           /'___\
      /\ \__/ /\ \__/  __  __  /\ \__/
      \ \ ,__\\ \ ,__\/\ \/\ \ \ \ ,__\
       \ \ \_/ \ \ \_/\ \ \_\ \ \ \ \_/
        \ \_\   \ \_\  \ \____/  \ \_\
         \/_/    \/_/   \/___/    \/_/

      v1.1.0
------------------------------------------------

 :: Method           : GET
 :: URL              : http://192.168.0.47/FUZZ
 :: Wordlist         : FUZZ: docs/vulnerabilities/fuzzer_wordlist.txt
 :: Follow redirects : false
 :: Calibration      : false
 :: Timeout          : 10
 :: Threads          : 40
 :: Matcher          : Response status: 200,204,301,302,307,401,403
------------------------------------------------

home.php                [Status: 302, Size: 0, Words: 1, Lines: 1]
profile.php             [Status: 200, Size: 4966, Words: 1081, Lines: 126]
README.md               [Status: 200, Size: 3983, Words: 557, Lines: 82]
api                     [Status: 301, Size: 350, Words: 21, Lines: 10]
panier.php              [Status: 200, Size: 6089, Words: 1887, Lines: 220]
logout.php              [Status: 302, Size: 0, Words: 1, Lines: 1]
ajouter-panier.php      [Status: 302, Size: 0, Words: 1, Lines: 1]
admin                   [Status: 301, Size: 352, Words: 21, Lines: 10]
menu.php                [Status: 200, Size: 3162, Words: 759, Lines: 96]
index.php               [Status: 200, Size: 3118, Words: 1173, Lines: 122]
test.php                [Status: 200, Size: 77620, Words: 3768, Lines: 889]
composer.lock           [Status: 200, Size: 3845, Words: 1377, Lines: 99]
config.php              [Status: 200, Size: 0, Words: 1, Lines: 1]
composer.json           [Status: 200, Size: 67, Words: 19, Lines: 6]
:: Progress: [27/27] :: Job [1/1] :: 6 req/sec :: Duration: [0:00:04] :: Errors: 0 ::
ergo@ergo-HP-Laptop-15-dw2xxx:~/Desktop/sec_lab_v2_final/seclab-soc-portfolio$ ffuf -w docs/vulnerabilities/fuzzer_wor
dlist.txt -u http://192.168.0.47/FUZZ
```

Figure 2: Directory fuzzing exposing sensitive paths like `/admin/` and `/api/`.

2 Phase 2: Initial Access (SQL Injection)

The attacker first navigated to the main landing page of the application (`index.html` / `index.php`), which presented the primary login portal. Utilizing a classic boolean-based SQL injection payload (`'or 1=1 -- -`) within the email input field, the authentication mechanism was entirely bypassed.

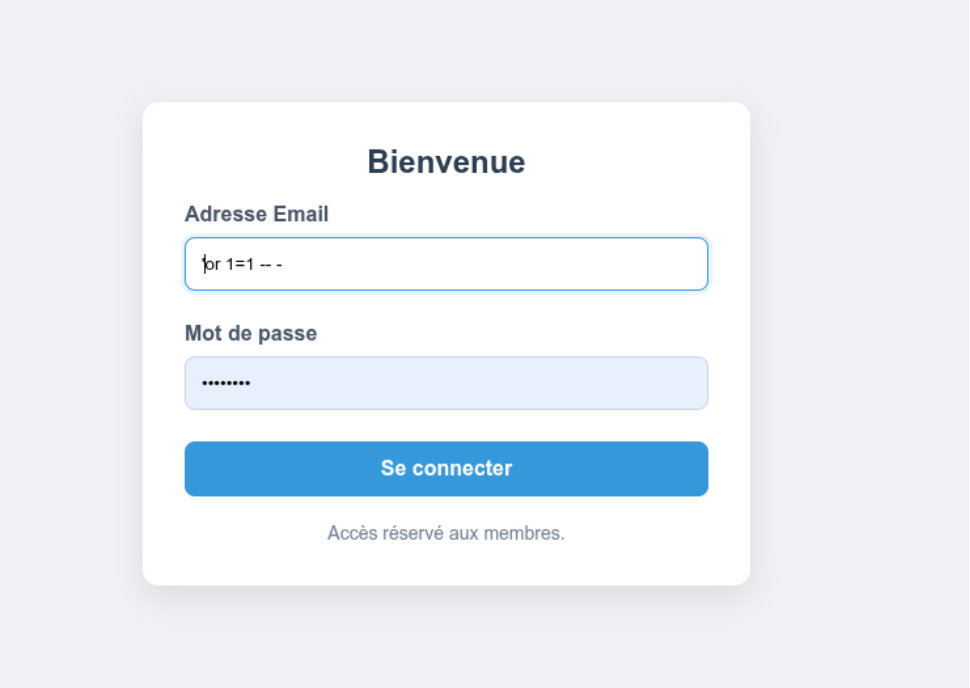


Figure 3: Bypassing authentication via SQL injection on the index page login portal.

3 Phase 3: Exploitation and Remote Code Execution (RCE)

Once authenticated, the attacker navigated to the restricted admin page, locating an internal administrative panel labeled "Diagnostic Serveur". This utility, meant for server debugging, was identified as vulnerable to OS Command Injection due to improper sanitization of semicolons (`;`).

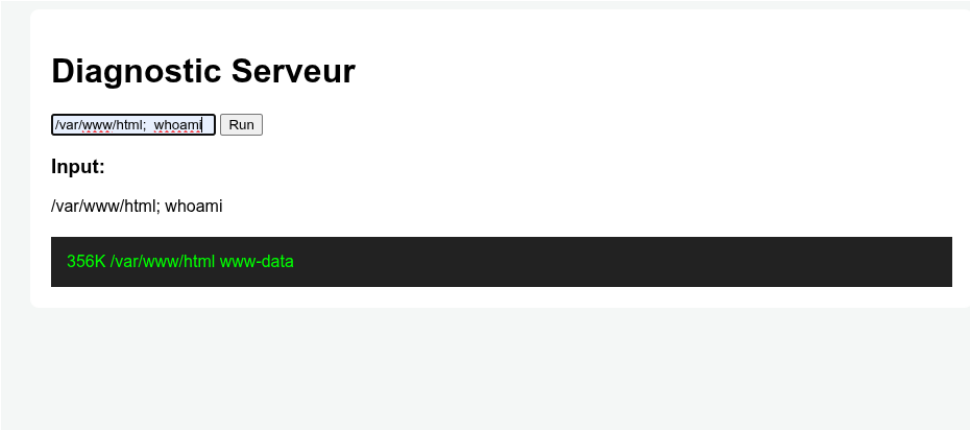


Figure 4: Execution of `whoami` via the admin Diagnostic Interface returning `www-data`.

With execution capabilities confirmed, the attacker constructed an intricate Python 3 reverse shell script to launch a persistent connection. Figure 5 displays the exact script utilized for the payload.

Figure 5: Close-up of the initial Python 3 reverse shell script crafted for the payload.

The script was then delivered via the vulnerable admin interface.

Figure 6: Delivery of the Python 3 reverse shell payload.

The shell was successfully caught by the active Netcat listener on the attacking machine, granting interactive lower-privileged (`www-data`) terminal access to the web server.

Figure 7: Interactive reverse shell session established.

4 Phase 4: Privilege Escalation and Objectives

After obtaining the initial shell, the attacker delivered deeply obfuscated exploit tooling, notably a Python script named `Ergo.py`. This script leverages a known kernel methodology (CopyFAIL / similar exploit structures).

Figure 8: Content of `Ergo.py` displaying the highly obfuscated CopyFAIL privilege escalation script.

Executing the script manipulated standard system constructs to launch a detached sub-command requesting an escalation to Superuser.

Figure 9: Privilege Escalation from `www-data` to `root` (uid=0) via `Ergo.py`.

The attacker now possessed full system control, confirmed by extracting the highly protected `/etc/shadow` file containing system credential hashes.

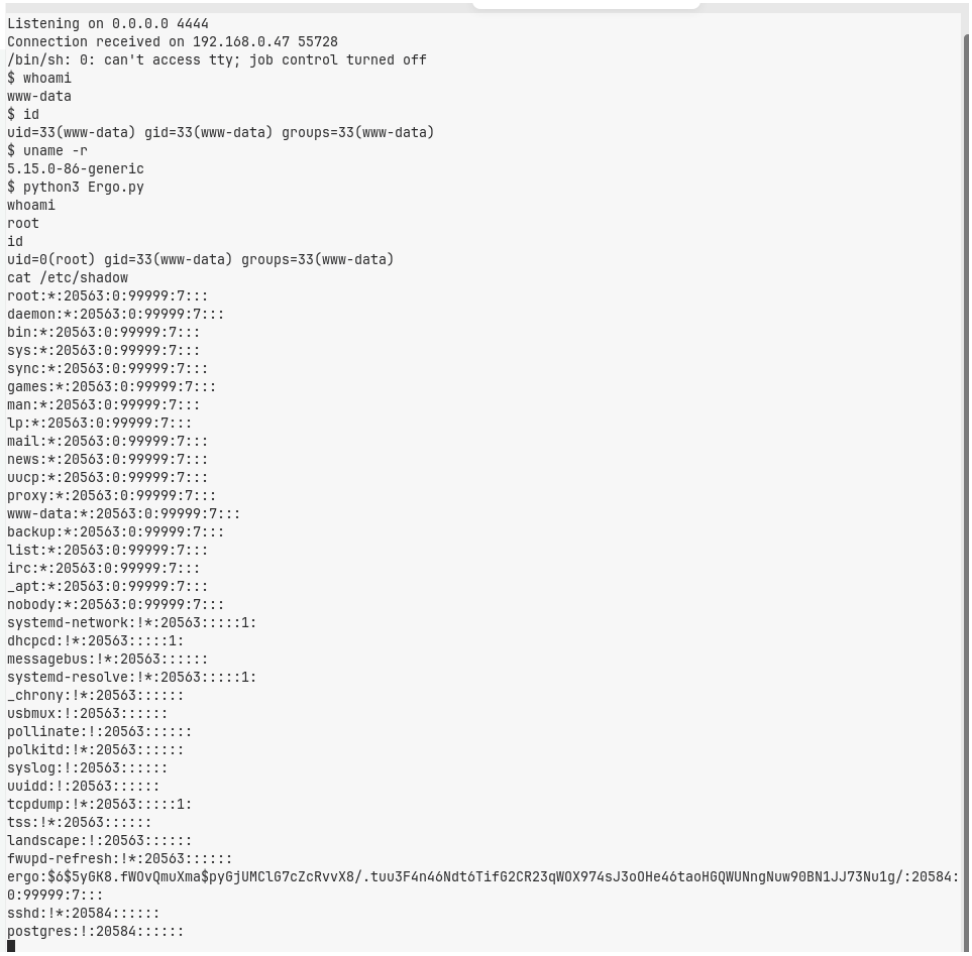


Figure 10: Extracting the `/etc/shadow` file indicating a full system compromise.

5 Phase 5: Implementation of Defense in Depth (AppArmor)

To rectify this architectural weakness, the administrator implemented and activated Mandatory Access Control profiles via AppArmor.

```
ergo@ergoweb:~$ sudo cat /etc/apparmor.d/restricted-shell
profile restricted-shell {
  /bin/sh mr,
  /usr/bin/dash mr,

  # dynamic linker — required for all ELF binaries
  /etc/ld.so.cache r,

  # allowed tools
  /usr/bin/du ix,
  /usr/lib/cargo/bin/coreutils/du ix,
  /usr/lib/cargo/bin/coreutils/ls ix,
  /usr/lib/cargo/bin/coreutils/cat ix,
 #/usr/bin/cat ix,
 #/usr/bin/ls ix,

  /proc/*/auxv r,
  /proc/*/status r,
  /proc/*/maps r,
  /usr/share/coreutils/locales/uucore/** r,

  # essential libraries
  /usr/lib/x86_64-linux-gnu/ld-linux-x86-64.so.2 mr,
  /usr/lib/x86_64-linux-gnu/libc.so.6 mr,
  /usr/lib/x86_64-linux-gnu/libselinux.so.1 mr,
  /usr/lib/x86_64-linux-gnu/** mr,
  /usr/lib/** mr,

  /var/www/** r,
  /tmp/** r,
  /var/log/apache2/error.log a,
  /var/log/apache2/access.log a,

  deny /etc/shadow r,
  deny /root/** rwx,
  deny /home/** rwx,
  deny /boot/** rwx,
  deny /bin/bash x,
  deny /usr/bin/bash x,
  audit deny /usr/bin/python* x,
  audit deny /usr/bin/python3.14 x,
  deny /usr/bin/curl x,
  deny /usr/bin/wget x,
  deny /usr/bin/nc x,
}
ergo@ergoweb:~$
```

Figure 11: The `restricted-shell` Profile explicitly denying access to system directories and executable abstractions (`python`, `curl`, `bash`).

These rules were systematically loaded onto the Apache processes, enforcing strict confinement boundaries on the application.

```
ergo@ergoweb:~$ sudo apparmor_parser -r /etc/apparmor.d/restricted-shell
sudo apparmor_parser -r /etc/apparmor.d/usr.sbin.apache2
ergo@ergoweb:~$ sudo systemctl restart apache2
ergo@ergoweb:~$ sudo aa-status | grep apach
   /usr/sbin/apache2
   /usr/sbin/apache2 (2410)
   /usr/sbin/apache2 (2413)
   /usr/sbin/apache2 (2414)
   /usr/sbin/apache2 (2416)
   /usr/sbin/apache2 (2417)
   /usr/sbin/apache2 (2421)
ergo@ergoweb:~$
```

Figure 12: Verifying the application of the AppArmor execution profiles against Apache.

6 Phase 6: Validation against Repeated Exploitation

With the confinement policies active, the exact same attack vectors were repeated through the web interface to assess mitigation efficiency.

When injecting the `whoami` command or the Python reverse shell payload, the underlying kernel now intercept and immediately rejected the sub-process spawning requests.

Both payloads returned `Permission denied` directly in the web application output instead of executing.

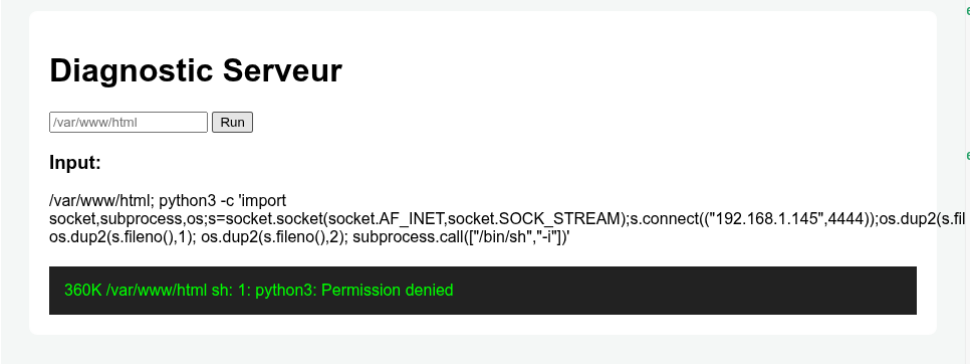


Figure 13: The reverse shell payload fails due to an OS-level permission violation against Python 3.

The security mitigation was positively confirmed via the kernel's audit ring buffer (`dmesg`), generating explicit `apparmor="DENIED"` alerts linked directly to the application trying to step outside its permitted execution boundaries.

Figure 14: Kernel audit logs explicitly exposing AppArmor denying the execution of restricted binaries.

7 Conclusion

The exercise fundamentally demonstrates the lethal progression of web application vulnerabilities when untethered. It highlights how minor application faults can rapidly pivot into total infrastructure compromise. Conversely, the introduction of AppArmor underscores the indispensability of Defense-in-Depth; by shifting security from the vulnerable application layer to the impassable kernel layer, the Remote Code Execution flaw was rendered entirely inert.